TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

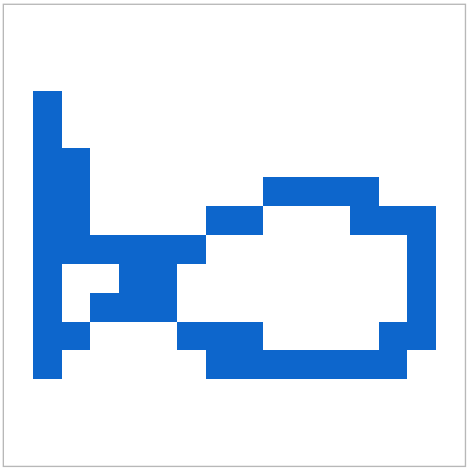
Mia and a big fish went to a sea. Lily wanted to see the sea. She pushed the button with it. But the printer mixed the seed inside. “What easy?” her mom asked. The seed was out quiet and did not feel neat. Liza picked it up and held it up the seed. “This is a fish!” he said. Then Mia walked into the room. She saw the paper and crossed the button. She pulled it out of the printer. It was not looking. M…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . .
1 . . . . . .
5 2 . . . . . .
5 a . 1 6 c b 2
5 c f 8 . . . a
5 6 c 3 2 . 1 a
4 . . 4 c c c .
. . . . . .
```



TONER-1-V’s drawing of a fish: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: “fish”, word 5 of 85. Class token 21 (fish).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K printer-tuned				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	80.6	6	0.8	64	256	6	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).